EWB CLOUD LABS | STUDENT PROJECT REPORT

AWS VPC Peering

Connect two VPCs. Prove private connectivity.

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Batch	AWS Batch1
Project	Project 01: VPC Peering (AWS Architect Associate)
Region	Asia Pacific (Mumbai) / ap-south-1
Build dates	21 October 2026 (network and instances), 23 October 2026 (peering, tests, cleanup)
Completion date	23 October 2026

Two VPCs (10.10.0.0/16 and 10.20.0.0/16) are connected with VPC peering. One public bastion in VPC A gives SSH access, using ProxyJump, to private EC2 instances in both VPCs. App-A and App-B have no public IP. All resources were deleted after the lab.

1. Architecture and resource inventory

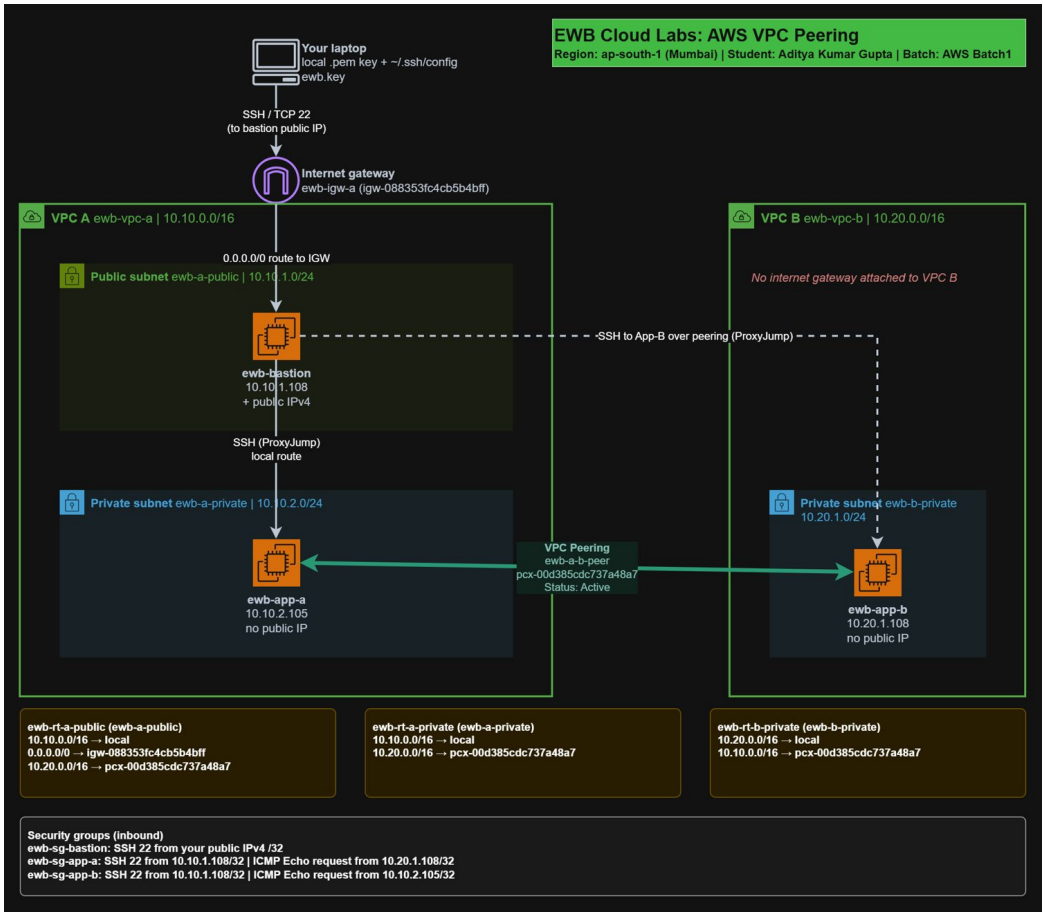


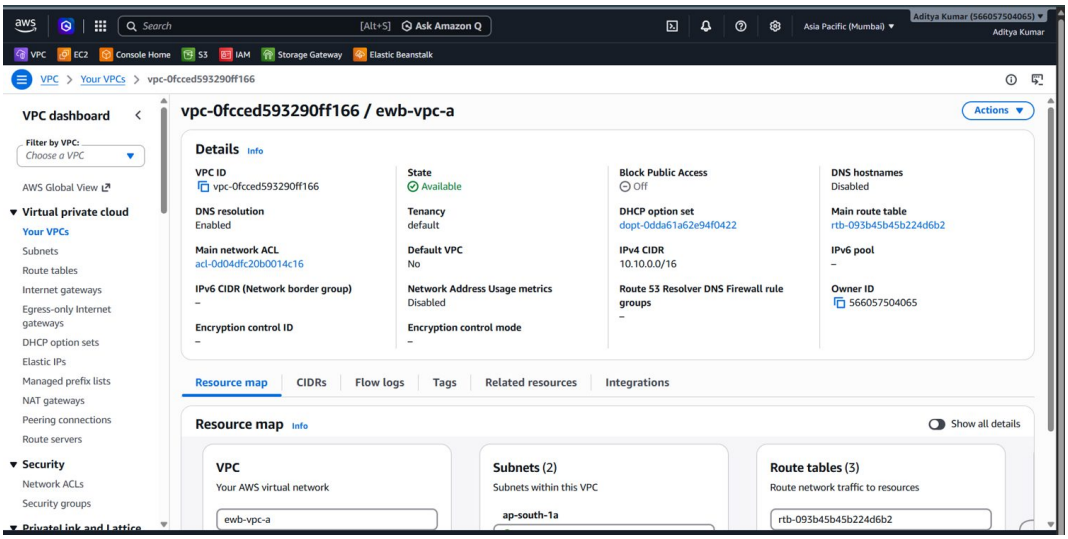
Figure 1: Final architecture with the actual resource IDs, private IPs, route tables and security group rules.

Resource inventory

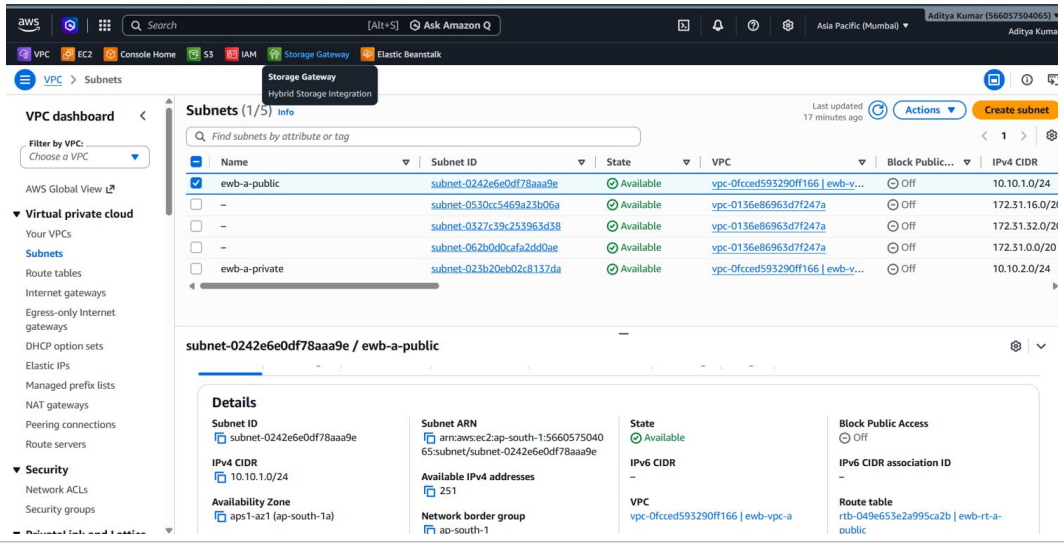
Resource	Name	ID	CIDR / IP
VPC A	ewb-vpc-a	vpc-0fcced593290ff166	10.10.0.0/16
Public subnet	ewb-a-public	subnet-0242e6e0df78aaa9e	10.10.1.0/24
Private subnet	ewb-a-private	subnet-023b20eb02c8137da	10.10.2.0/24
Internet gateway	ewb-igw-a	igw-088353fc4cb5b4bff	attached to VPC A
VPC B	ewb-vpc-b	vpc-032ad1b0062f2c9ce	10.20.0.0/16
Private subnet	ewb-b-private	subnet-0399e45a6a15ea5a2	10.20.1.0/24
Peering connection	ewb-a-b-peer	pcx-00d385cdc737a48a7	VPC A to VPC B
Instance	ewb-bastion	i-0cd27abfec699fa83	10.10.1.108 + public IPv4
Instance	ewb-app-a	i-0c9ab388f415eca9e	10.10.2.105, no public IP
Instance	ewb-app-b	i-057a7ebb8e0ae2bba	10.20.1.108, no public IP

Private IPs are AWS-assigned (the workbook allows this when every rule uses the actual values), so the security group rules and tests use 10.10.1.108, 10.10.2.105 and 10.20.1.108. Instances are Amazon Linux 2023, t3.micro, in ap-south-1a.

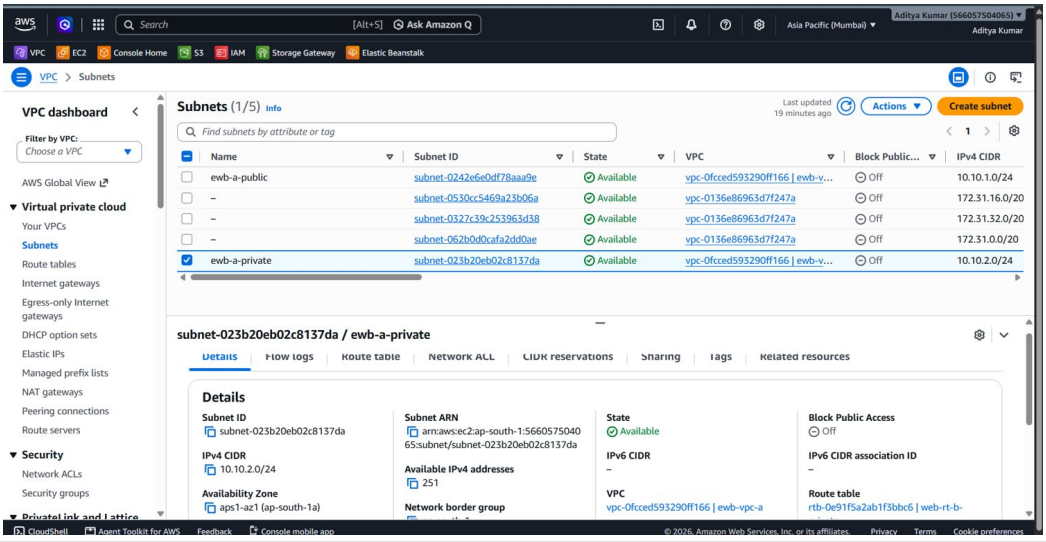
2. VPCs, subnets and internet gateway



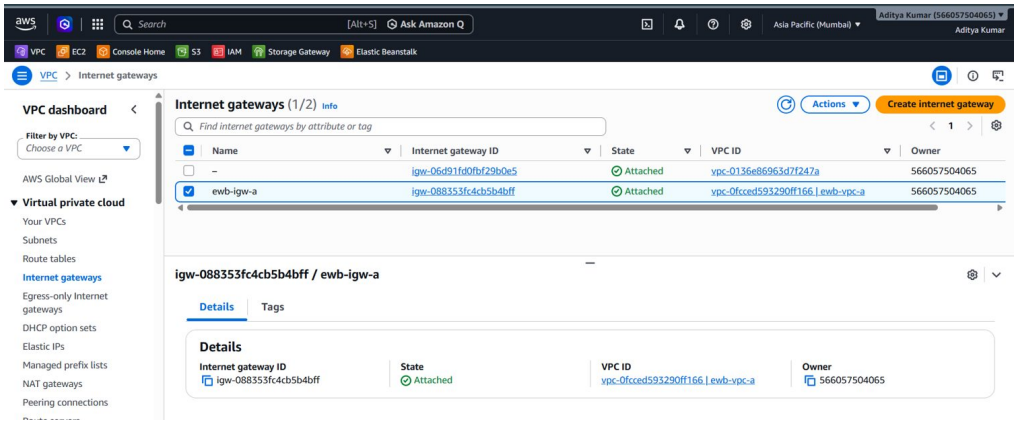
Laptop console: ewb-vpc-a, vpc-0fcced593290ff166, CIDR 10.10.0.0/16, Available.



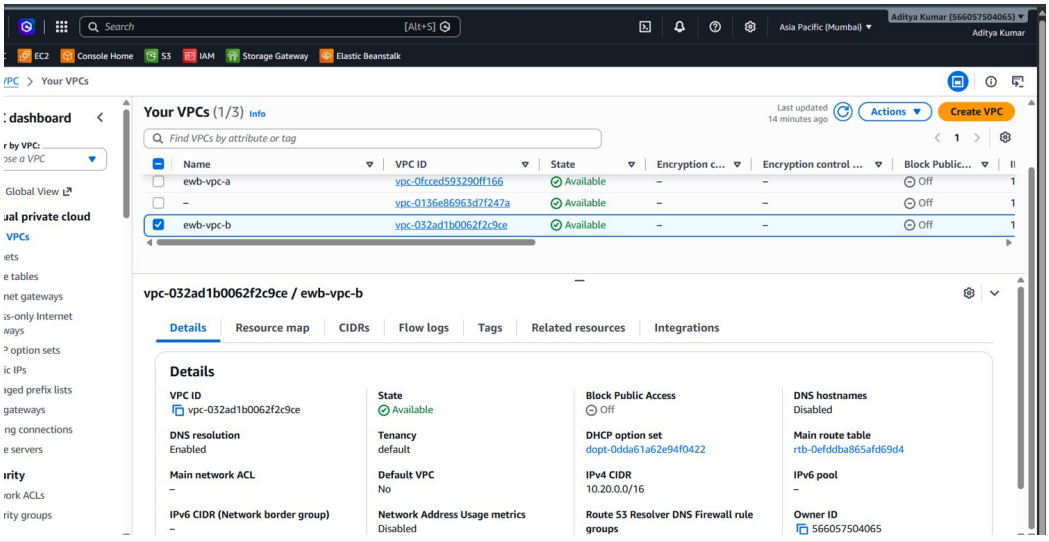
Console: ewb-a-public (10.10.1.0/24) in ewb-vpc-a, ap-south-1a.



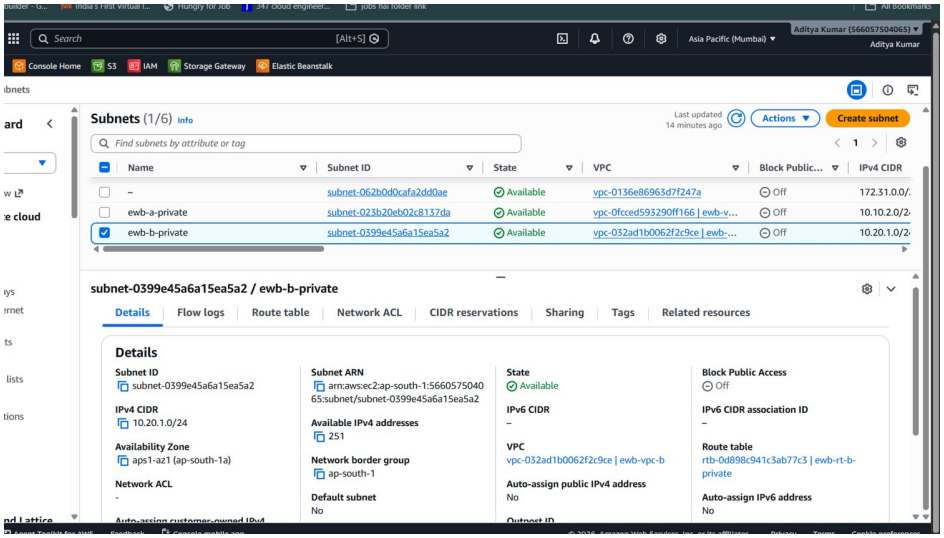
Console: ewb-a-private (10.10.2.0/24) in ewb-vpc-a, ap-south-1a.



Console: ewb-igw-a attached to ewb-vpc-a (state Attached).



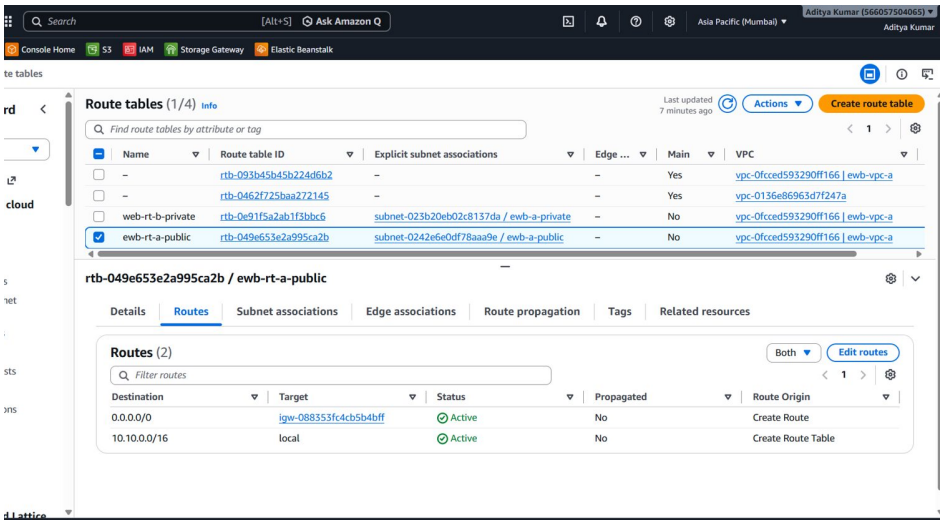
Console: ewb-vpc-b, vpc-032ad1b0062f2c9ce, CIDR 10.20.0.0/16.



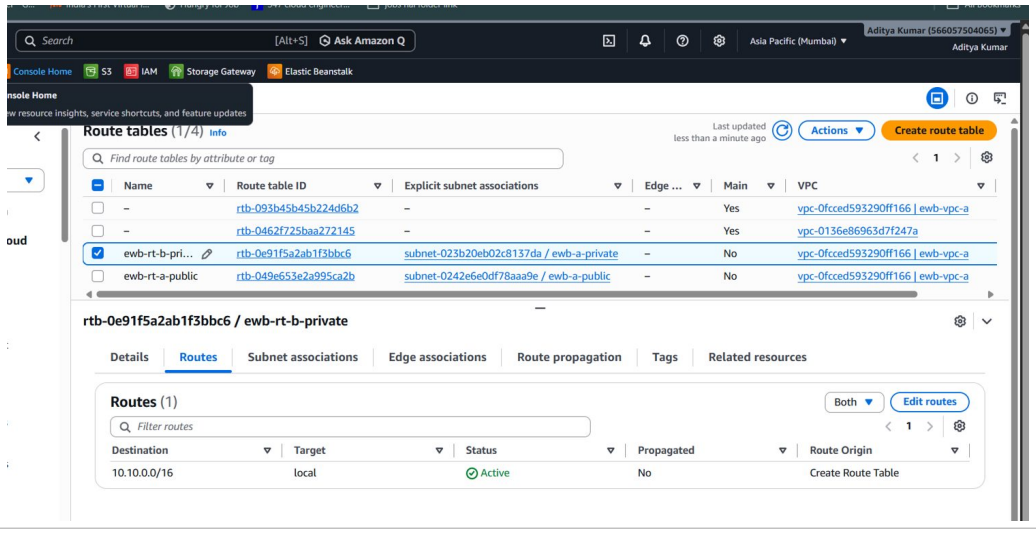
Console: ewb-b-private (10.20.1.0/24) in ewb-vpc-b, auto-assign public IPv4 = No.

3. Route tables and subnet associations

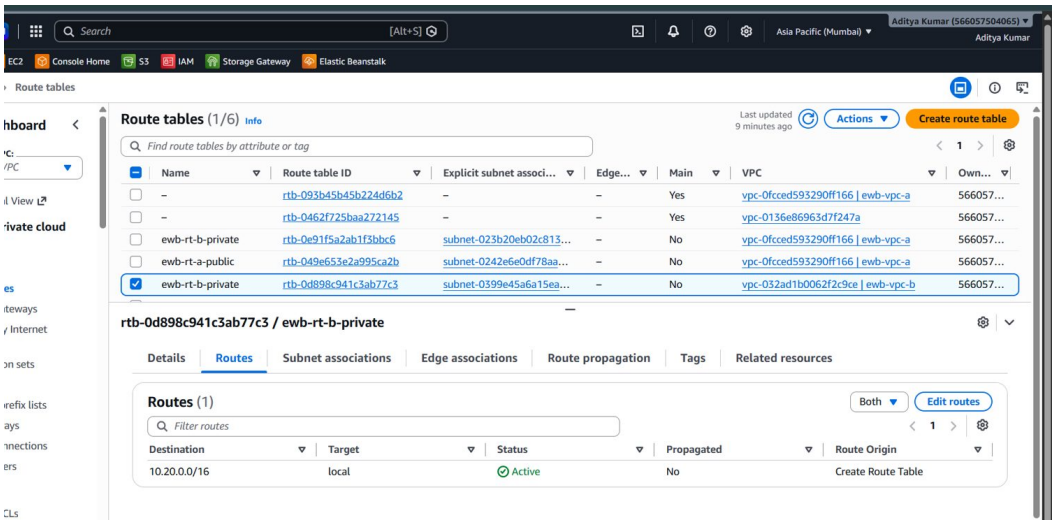
Before peering



ewb-rt-a-public before peering: 0.0.0.0/0 to igw-088353fc4cb5b4bff and the local route; associated with ewb-a-public.



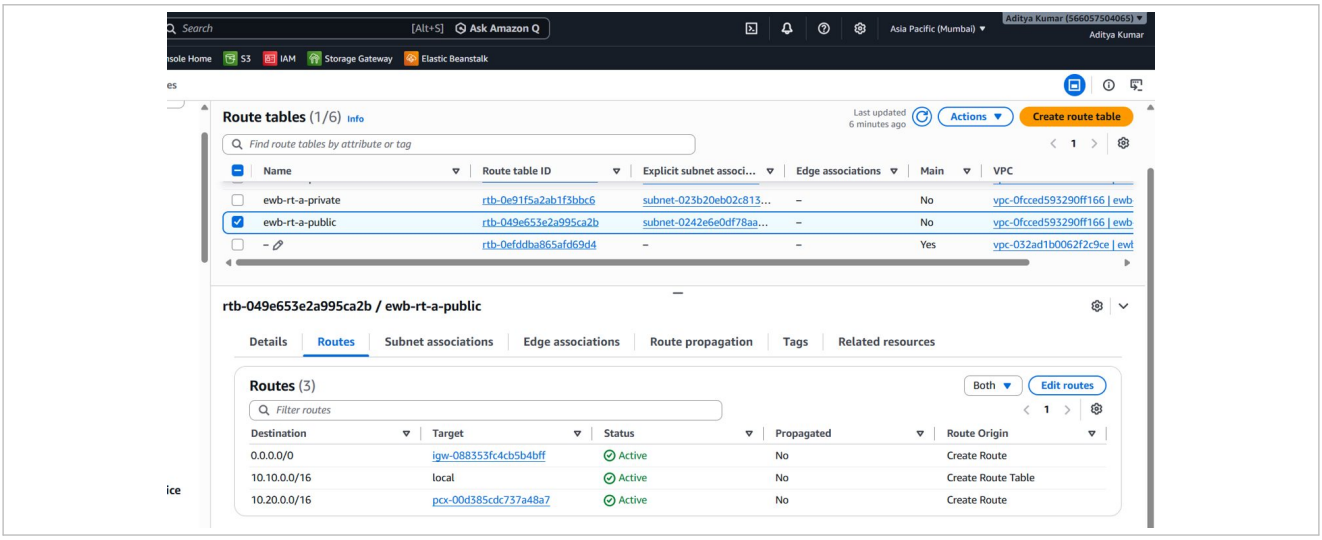
A-private route table before peering: local route only; associated with ewb-a-private. (In this early screenshot the table still carried a wrong name; it was renamed ewb-rt-a-private, same ID rtb-0e91f5a2ab1f3bbc6.)



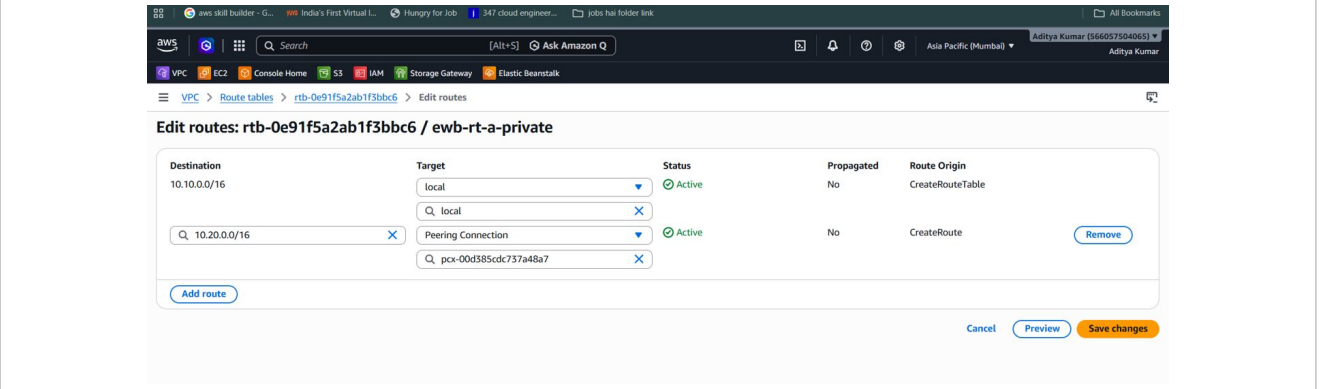
ewb-rt-b-private before peering: local route 10.20.0.0/16 only; associated with ewb-b-private.

After peering (final routes)

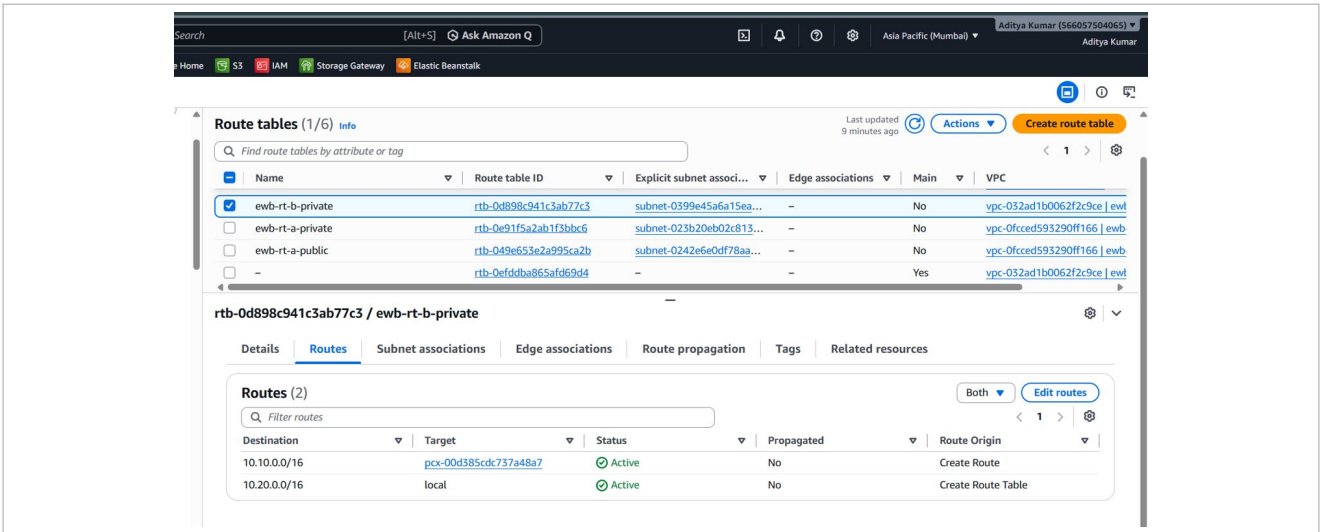
Route table	Destination	Target	Purpose
ewb-rt-a-public	10.10.0.0/16 / 0.0.0.0/0 / 10.20.0.0/16	local / igw-088353... / pcx-00d385...	Bastion to App-A, internet, bastion SSH to App-B
ewb-rt-a-private	10.10.0.0/16 / 10.20.0.0/16	local / pcx-00d385cdc737a48a7	Forward path App-A to VPC B
ewb-rt-b-private	10.20.0.0/16 / 10.10.0.0/16	local / pcx-00d385cdc737a48a7	Return path to VPC A



ewb-rt-a-public final: three routes (internet gateway, local, peering) and its explicit subnet association.

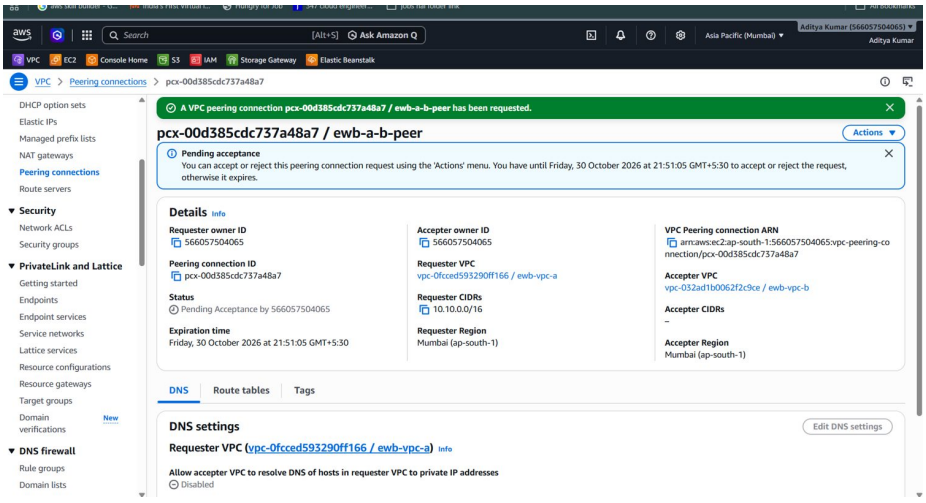


ewb-rt-a-private final (edit routes page): 10.10.0.0/16 local and 10.20.0.0/16 to pcx-00d385cdc737a48a7.

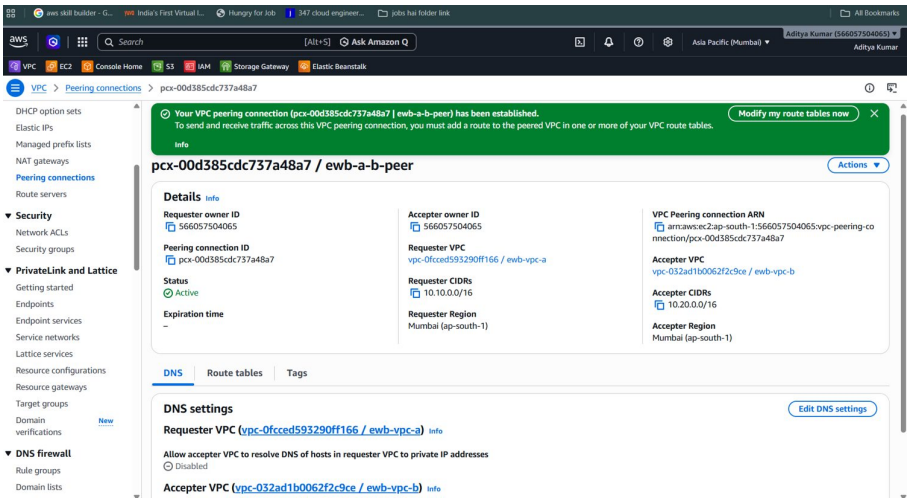


ewb-rt-b-private final: 10.10.0.0/16 to pcx-00d385cdc737a48a7 and local route; table list shows each explicit subnet association.

4. Peering connection, security groups and instances



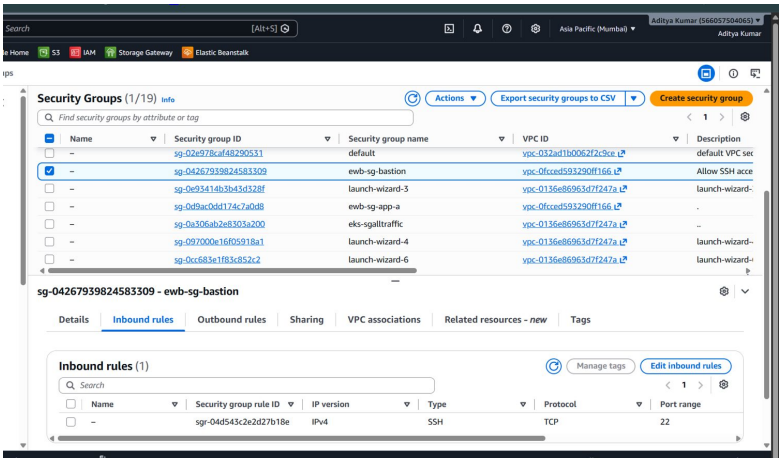
Peering ewb-a-b-peer (pcx-00d385cdc737a48a7) requested: Pending acceptance.



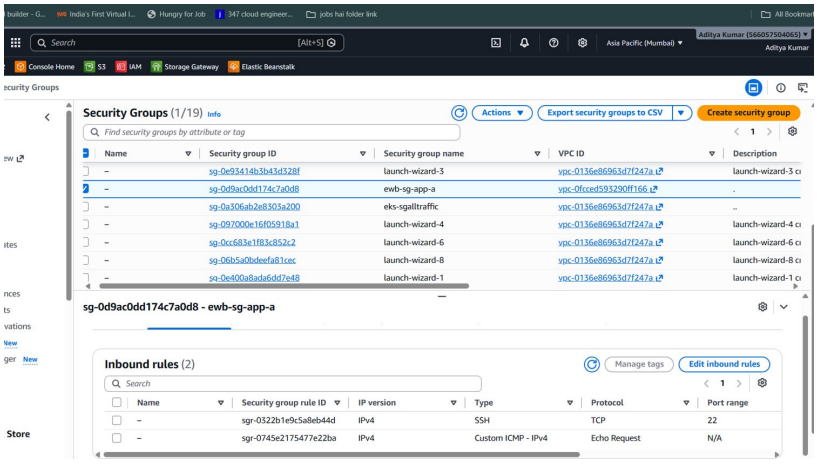
Peering accepted: status Active, requester ewb-vpc-a (10.10.0.0/16), accepter ewb-vpc-b (10.20.0.0/16).

Security group inbound rules

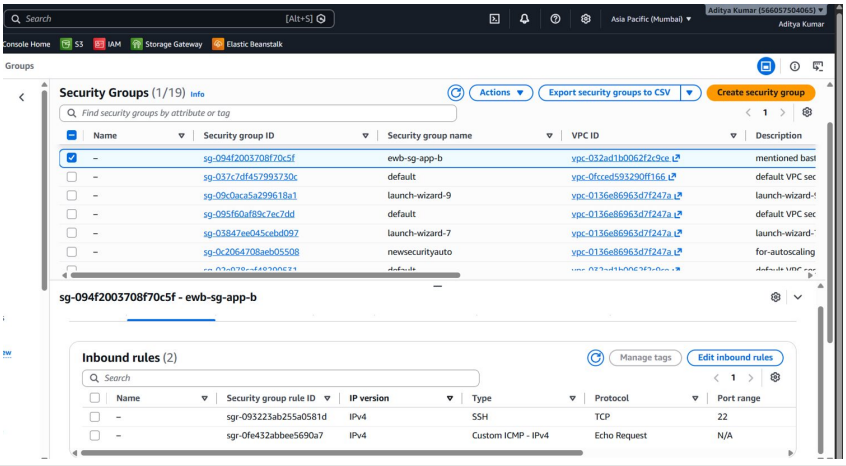
Security group	Type / port	Source
ewb-sg-bastion	SSH TCP 22	my public IPv4 /32
ewb-sg-app-a	SSH TCP 22	10.10.1.108/32 (bastion)
ewb-sg-app-a	Custom ICMP IPv4, Echo Request	10.20.1.108/32 (App-B)
ewb-sg-app-b	SSH TCP 22	10.10.1.108/32 (bastion)
ewb-sg-app-b	Custom ICMP IPv4, Echo Request	10.10.2.105/32 (App-A)



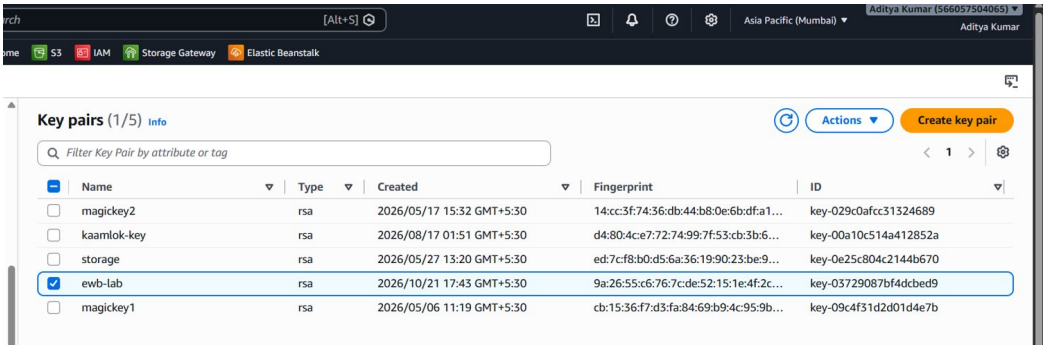
ewb-sg-bastion inbound rules: one SSH rule (source column is cut off by the window width).



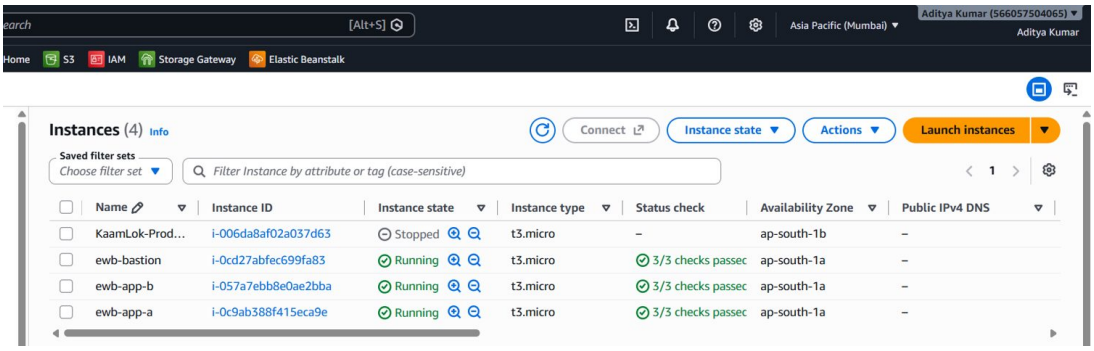
ewb-sg-app-a inbound rules: SSH and ICMP Echo Request.



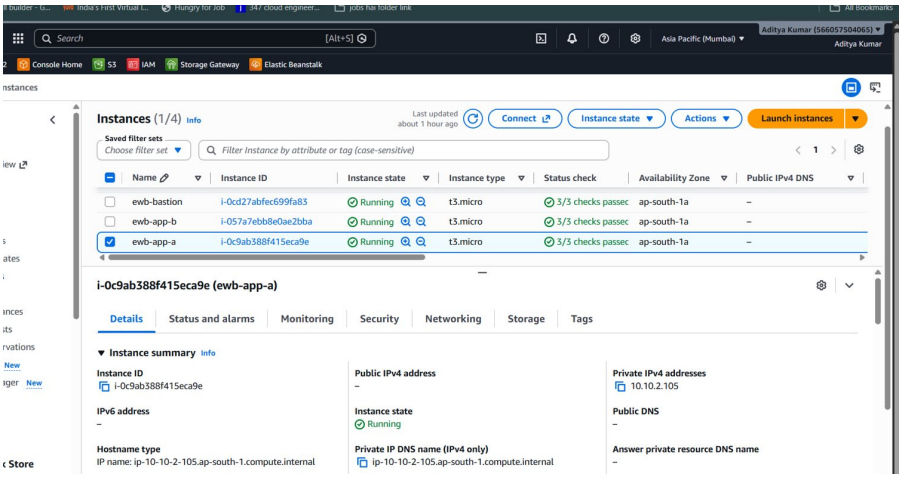
ewb-sg-app-b inbound rules: SSH and ICMP Echo Request (sources visible in screenshots 27 and 29).



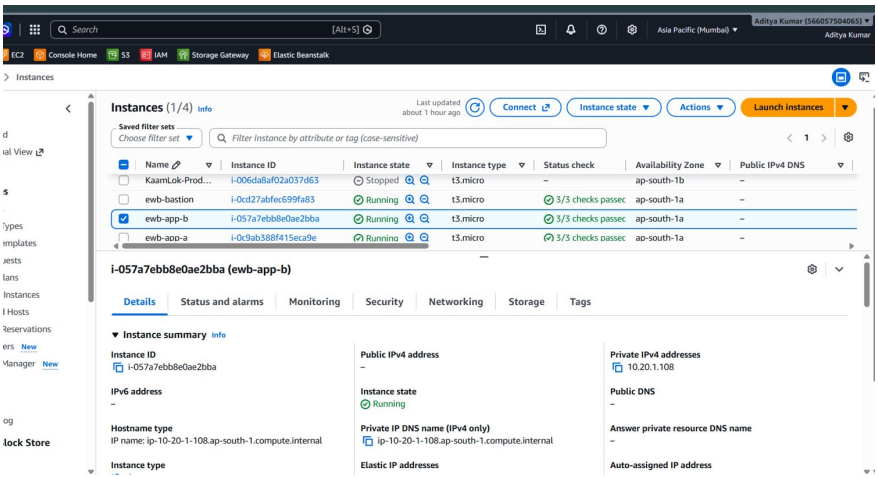
Key pair ewb-lab (RSA). The private key stays on my laptop and is not in this report or repository.



Instances ewb-bastion, ewb-app-b and ewb-app-a: Running, t3.micro, ap-south-1a (KaamLok-Prod is an unrelated stopped instance).



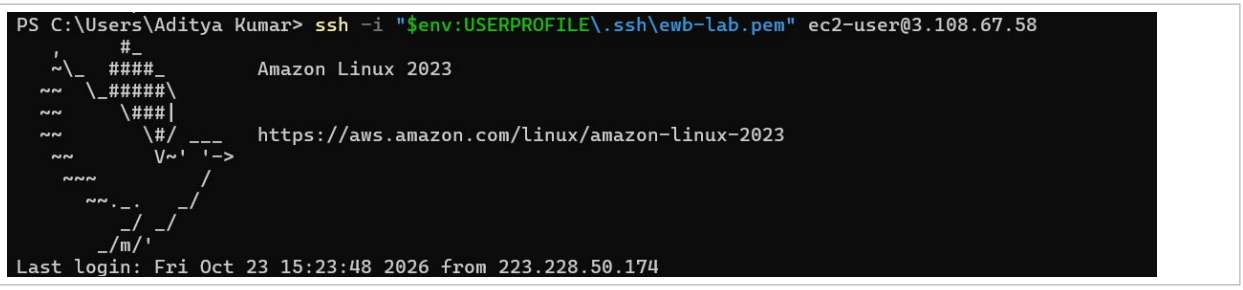
ewb-app-a: private IPv4 10.10.2.105, Public IPv4 address empty.



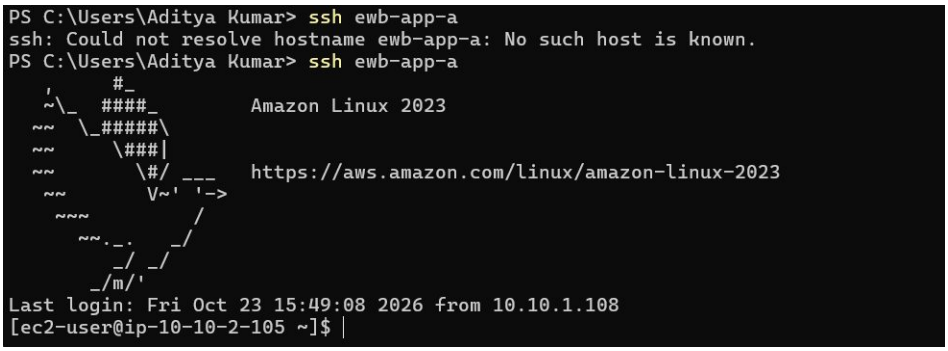
ewb-app-b: private IPv4 10.20.1.108, Public IPv4 address empty.

5. Test evidence

Test	Where / command	Expected	Actual
01 Baseline (before peering)	App-A: ping -c 4 10.20.1.108	No replies	100% packet loss
02 Forward path	App-A: ping -c 4 10.20.1.108	0% loss	4 received, 0% loss
03 Reverse path	App-B: ping -c 4 10.10.2.105	0% loss	4 received, 0% loss
04 Private SSH	Laptop: ssh ewb-app-b	Shell on App-B via bastion	Shell on ip-10-20-1-108
05 Isolation	EC2 details for App-A and App-B	No public IPv4	Public IPv4 empty on both



Laptop: ssh to the bastion public IP succeeded (before peering).



Laptop: ssh ewb-app-a through the bastion. The first attempt failed to resolve the host name, the second connected.

```
PS C:\Users\Aditya Kumar> ssh -G ewb-app-a | Select-String "hostname|user|identityfile|proxyjump"

user ec2-user
hostname 10.10.2.105
canonicalizehostname false
identityfile C:/Users/Aditya Kumar/.ssh/ewb-lab.pem
userknownhostsfile C:\Users\Aditya Kumar/.ssh/known_hosts C:\Users\Aditya Kumar/.ssh/known_hosts2
syslogfacility USER
proxyjump ewb-bastion


PS C:\Users\Aditya Kumar> ssh ewb-app-a
The authenticity of host '10.10.2.105 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:82sCTyfdA3+7nWxeR66N1B9Liudf/ldq39v0mlMjTcM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.105' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-105 ~]$ hostname
hostname -I
ping -c 4 10.20.1.108
ip-10-10-2-105.ap-south-1.compute.internal
10.10.2.105
PING 10.20.1.108 (10.20.1.108) 56(84) bytes of data.

--- 10.20.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms

[ec2-user@ip-10-10-2-105 ~]$ hostname
ip-10-10-2-105.ap-south-1.compute.internal
[ec2-user@ip-10-10-2-105 ~]$ hostname -I
10.10.2.105
[ec2-user@ip-10-10-2-105 ~]$ ping -c 4 10.20.1.108
PING 10.20.1.108 (10.20.1.108) 56(84) bytes of data.

--- 10.20.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3081ms
```

Test 01, App-A: baseline ping to 10.20.1.108 before peering and routes, 4 transmitted, 0 received, 100% packet loss. SSH config confirms ProxyJump through ewb-bastion.

```
--- 10.20.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3123ms

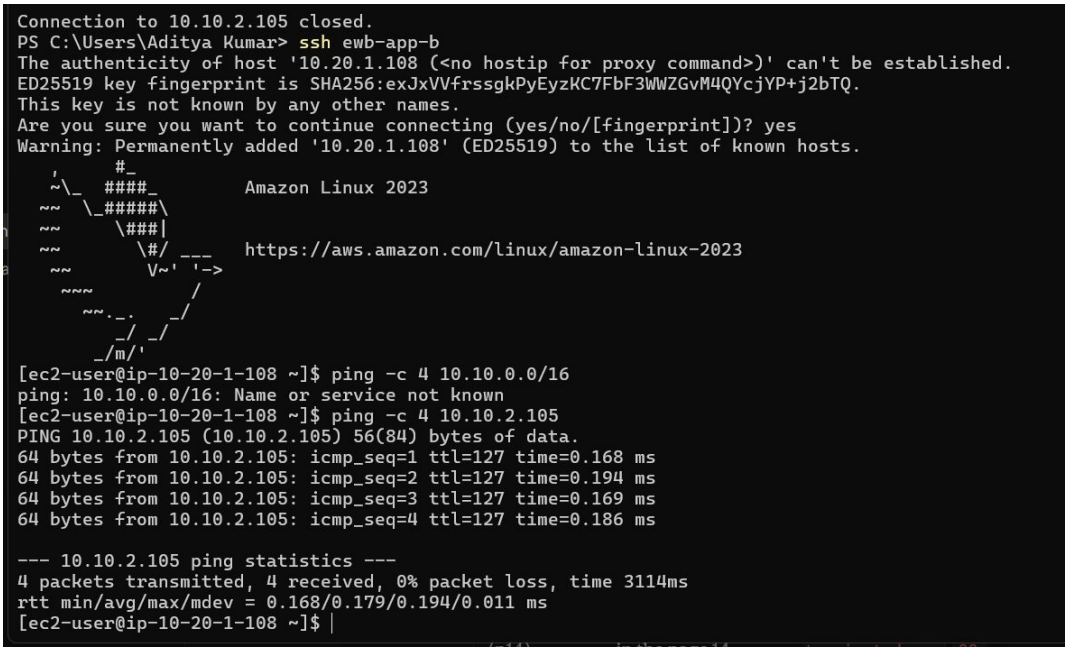
[ec2-user@ip-10-10-2-105 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
PS C:\Users\Aditya Kumar> ssh ewb-app-a
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 16:32:56 2026 from 10.10.1.108
[ec2-user@ip-10-10-2-105 ~]$ ping -c 4 10.10.1.108
PING 10.10.1.108 (10.10.1.108) 56(84) bytes of data.

--- 10.10.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3105ms

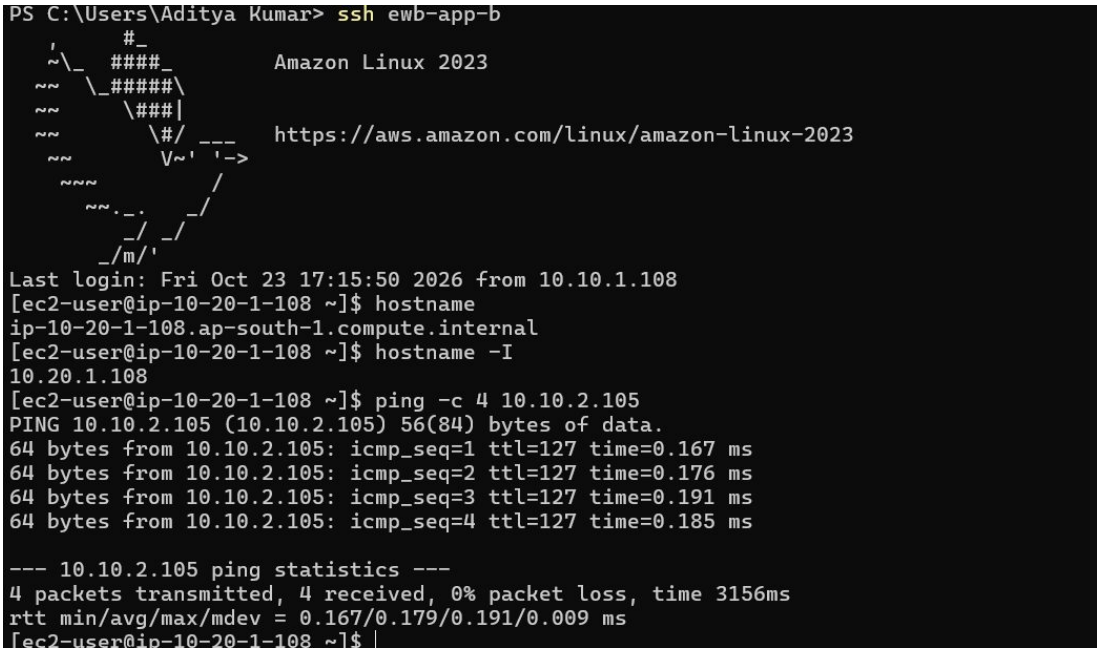
[ec2-user@ip-10-10-2-105 ~]$ ping -c 4 10.20.1.108
PING 10.20.1.108 (10.20.1.108) 56(84) bytes of data.
64 bytes from 10.20.1.108: icmp_seq=1 ttl=127 time=0.597 ms
64 bytes from 10.20.1.108: icmp_seq=2 ttl=127 time=0.163 ms
64 bytes from 10.20.1.108: icmp_seq=3 ttl=127 time=0.248 ms
64 bytes from 10.20.1.108: icmp_seq=4 ttl=127 time=0.183 ms

--- 10.20.1.108 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3106ms
rtt min/avg/max/mdev = 0.163/0.297/0.597/0.175 ms
```

Test 02, App-A: ping 10.20.1.108 after routes, 4 received, 0% packet loss. (The ping to the bastion private IP above it fails as expected: the bastion group allows SSH only.)



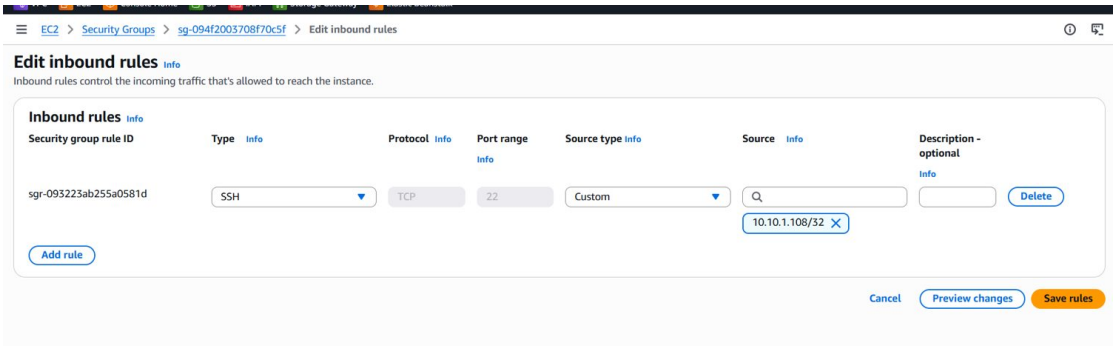
Test 03, App-B: ping 10.10.2.105, 4 received, 0% packet loss. (The first command, a ping to the CIDR 10.10.0.0/16, was a typo and was not a valid test.)



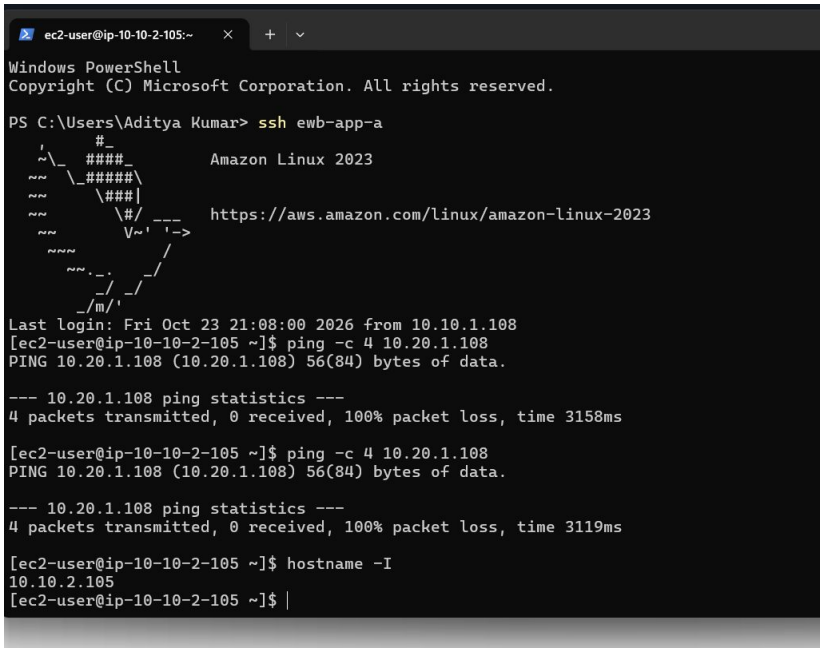
Test 04, laptop to App-B through the bastion: hostname ip-10-20-1-108, hostname -I 10.20.1.108, ping to 10.10.2.105 with 0% loss.

6. Controlled failure and recovery

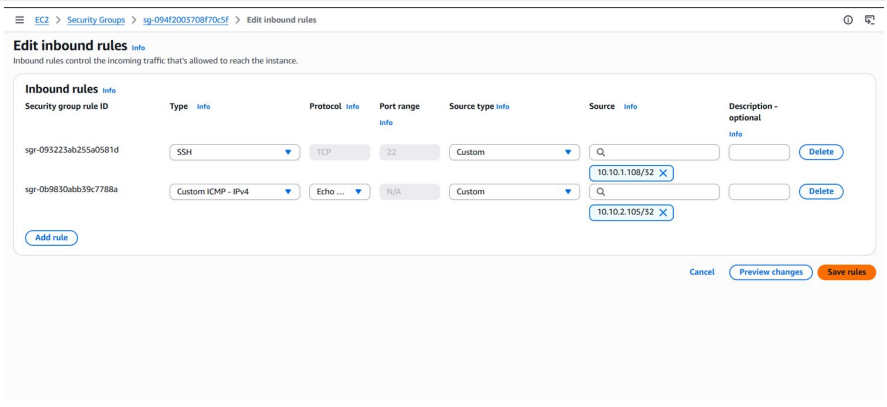
I removed only the ICMP Echo Request rule from ewb-sg-app-b and kept the SSH rule. The ping from App-A to App-B then failed even though the peering was Active and both routes were in place. Packets reached App-B, but its security group dropped them, so no replies came back. This shows that a route alone does not allow traffic: the destination security group must allow it too.



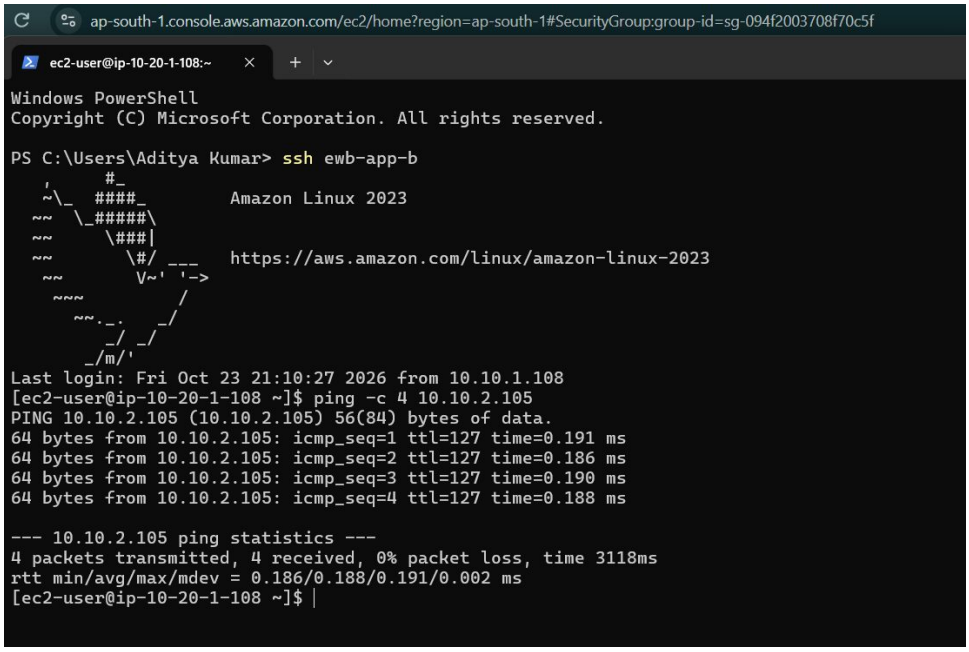
Step 1, ewb-sg-app-b: only the SSH rule from 10.10.1.108/32 remains (ICMP rule removed).



Step 2, App-A: ping -c 4 10.20.1.108 run twice, 4 transmitted, 0 received, 100% packet loss both times.



Step 3, ewb-sg-app-b: ICMP Echo Request rule restored from 10.10.2.105/32 next to the SSH rule.



Step 4, App-B: ping 10.10.2.105, 4 received, 0% packet loss. Note: this recovery ping was run from App-B to App-A; the original App-A to App-B direction was not re-captured after the rule was restored.

7. Forward path and return path

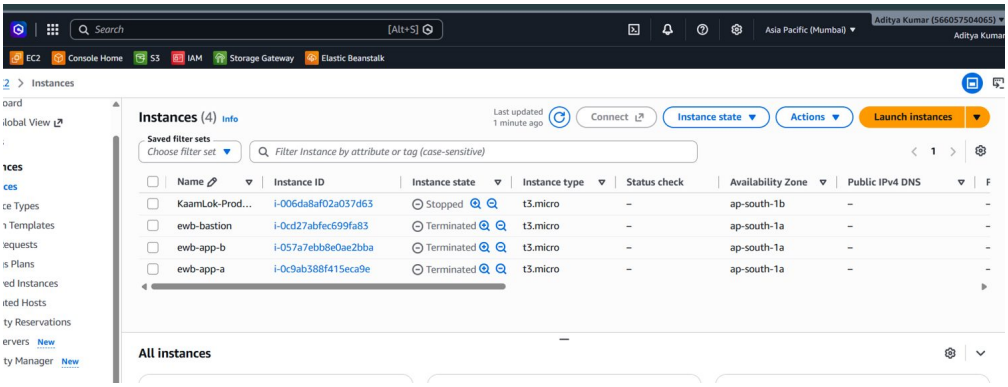
Forward path. App-A (10.10.2.105) pings App-B (10.20.1.108). The route table of ewb-a-private sends 10.20.0.0/16 to the peering connection pcx-00d385cdc737a48a7, and ewb-sg-app-b allows ICMP Echo Request from 10.10.2.105/32.

Return path. App-B replies to 10.10.2.105. The route table of ewb-b-private sends 10.10.0.0/16 to the same peering connection. Security groups are stateful, so the reply is allowed automatically. Without the return route the replies are dropped and the sender sees 100% loss.

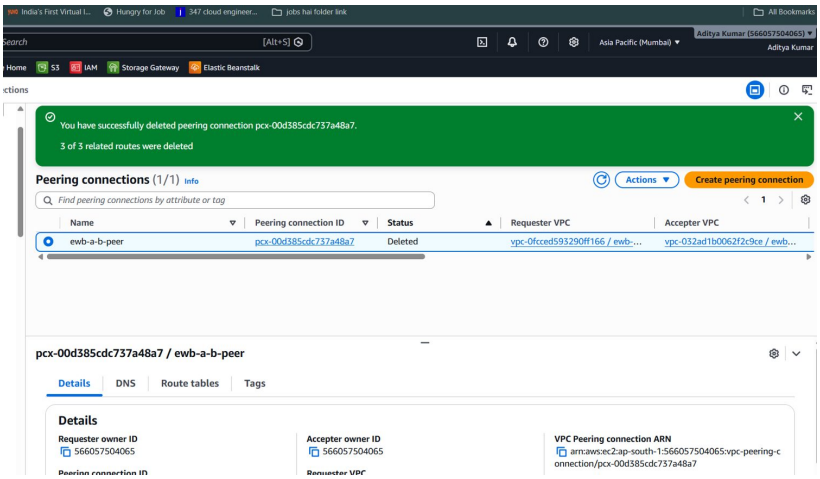
Bastion path. With ProxyJump the bastion (10.10.1.108) opens the SSH connection to App-B, so ewb-rt-a-public also needs 10.20.0.0/16 to the peering connection, and ewb-sg-app-b allows SSH from 10.10.1.108/32.

8. Cleanup

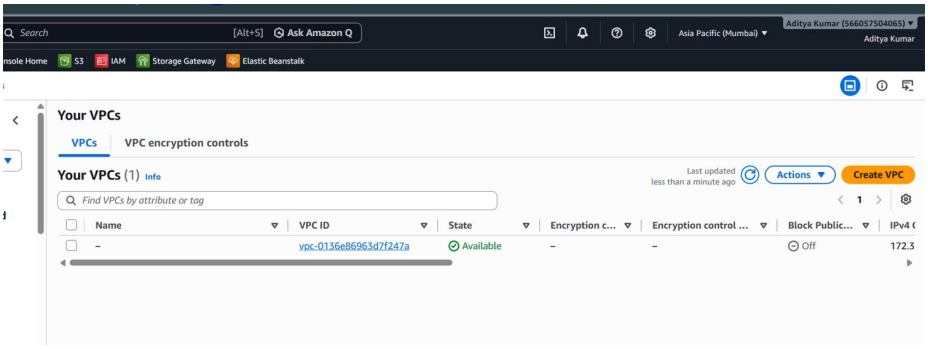
All lab resources were deleted on 23 October 2026 after the evidence was saved: instances terminated, peering connection deleted (3 of 3 related routes removed), and both lab VPCs deleted. Only the account default VPC remains.



Instances ewb-bastion, ewb-app-b and ewb-app-a: Terminated.



Peering connection pcx-00d385cdc737a48a7 deleted, 3 of 3 related routes deleted.



Your VPCs shows only the default VPC; ewb-vpc-a and ewb-vpc-b are gone.

9. What I learned

A public subnet needs an internet gateway route and the instance needs a public address; only the bastion had both. VPC peering is only a link: each side needs a route to the other VPC and the destination needs a security group rule. A bastion with SSH ProxyJump lets me manage private servers without copying the private key onto any instance. Rules scoped to /32 sources keep access limited to the lab hosts. Peering is non-transitive, so VPC B cannot use VPC A internet gateway.